Abdul, Jon – Simulation

Model C: Integrated Resilience Model

- **Scope:** Full Bavarian wood supply chain from forest to end market, testing resilience against multiple disruption types. This is the broadest option and matches your title best.

- **Method:** System Dynamics, AnyLogic.

- **Inputs:** Forest stock by species (spruce vs mixed), annual harvest, bark beetle rate, drought severity index, sawmill capacity, energy cost index, transport cost, labor availability, timber price, construction demand, pellet demand, government reforestation subsidies, import volumes from other regions.

- **Outputs:** Resilience indicators over time - recovery time after shock, price stability, supply continuity, inventory buffer adequacy. Compared across scenarios over 5-10 years.

- **Model structure:** 3 subsystems - forest supply (standing stock, harvest rate, reforestation pipeline, climate vulnerability), processing (sawmill capacity, energy costs, labor, production rate), market demand (construction, heating/pellets, exports).

  Key stocks: healthy forest, damaged forest, timber inventory, sawmill capacity. Multiple reinforcing and balancing loops connecting all three subsystems. Resilience measured through 3 dimensions similar to Ma et al. (2025): ecological (forest health, species diversity), economic (price stability, sawmill margins), infrastructure (transport, logistics, storage).

- **Scenarios:** (1) Baseline - normal operations, (2) Climate shock - severe drought + beetle outbreak, (3) Energy shock - oil/gas price spike, (4) Policy intervention - government funding for mixed species reforestation + energy transition subsidies.